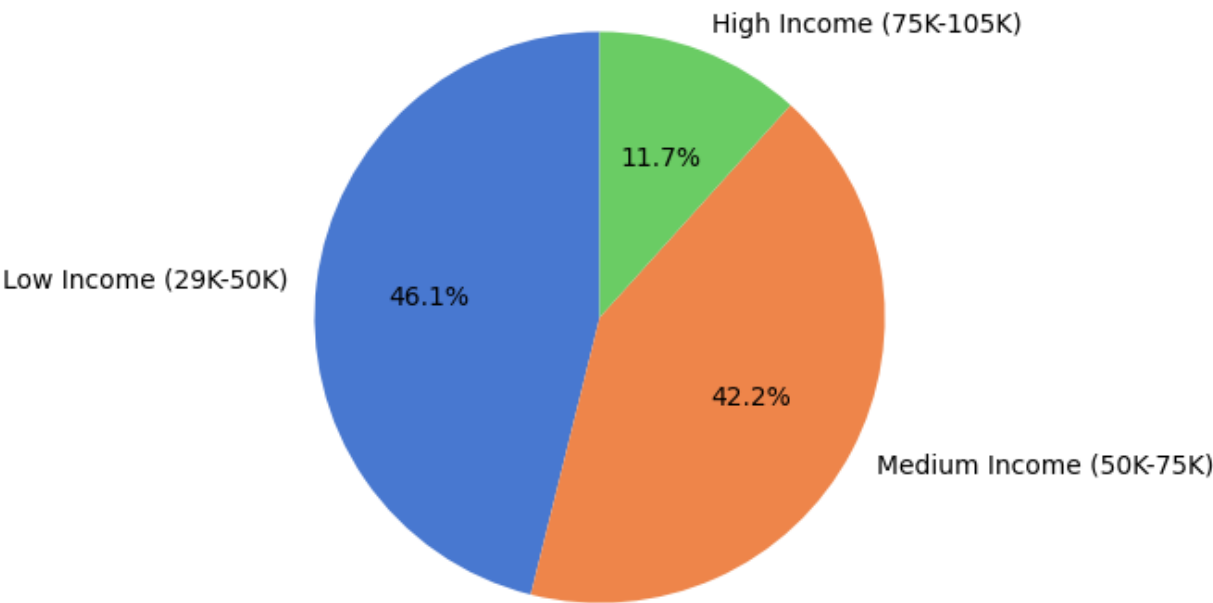
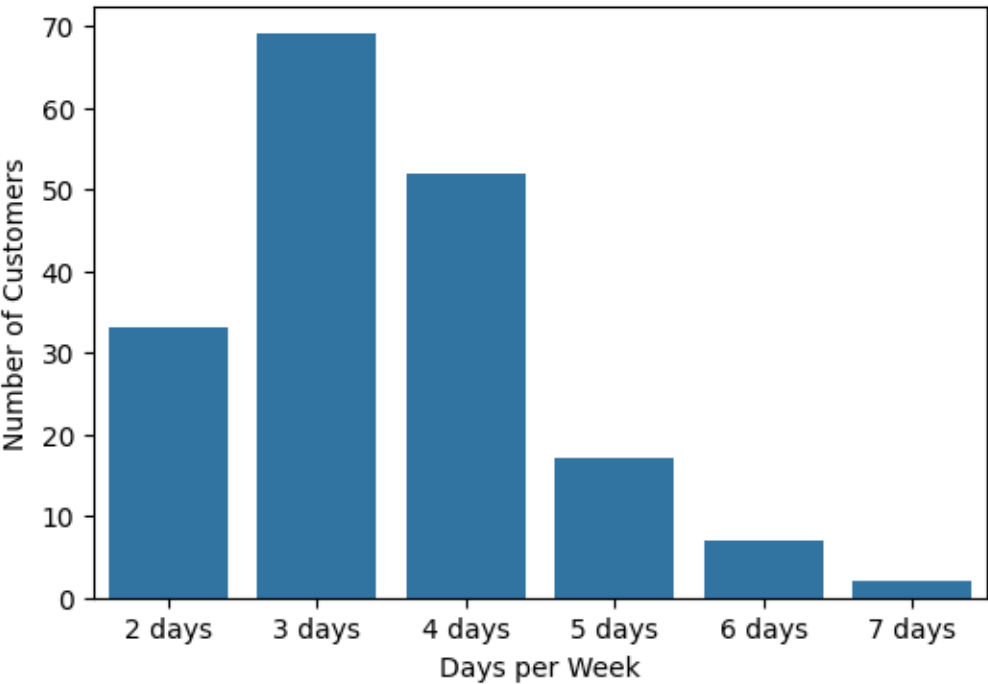


Aerofit: Customer Profiling & Product Insights

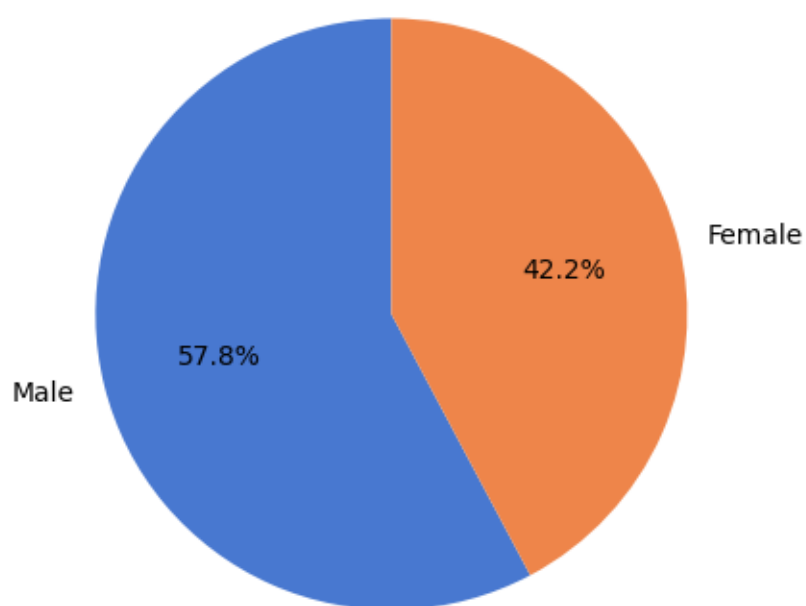
Income_Group Distribution



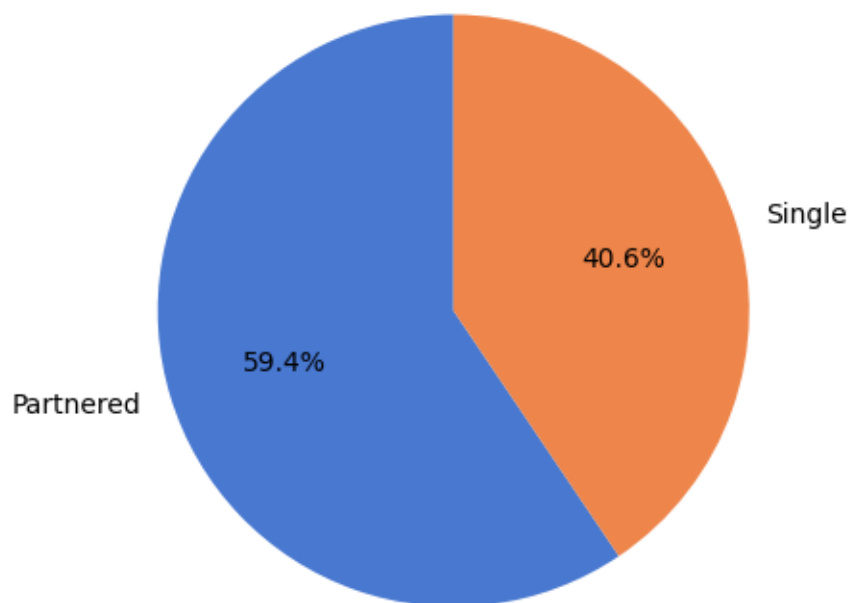
Usage per Week Distribution

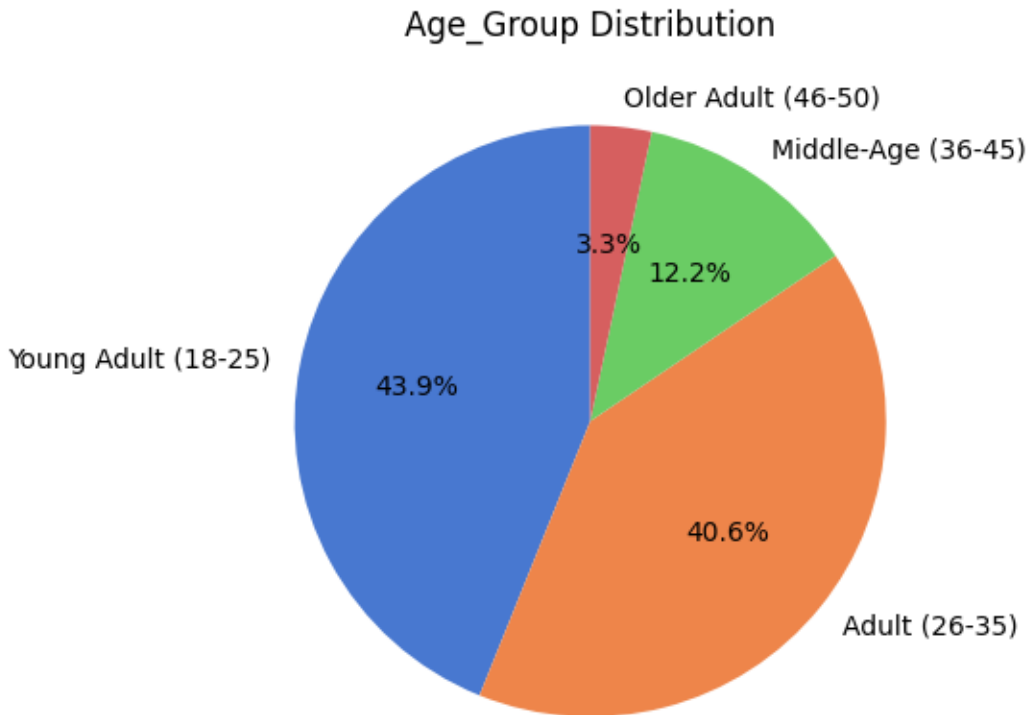


Gender Distribution



MaritalStatus Distribution





Univariate Analysis - Categorical Columns Quick Insights

Income Group

- Majority fall into Low to Medium Income (159 out of 180 customers, ~88%).
- Only a small fraction are High Income (21, 12%), suggesting most purchases come from middle-tier income customers.

Usage per Week

- Most customers plan to use the treadmill 2–4 days per week (~154, 86%), with the peak at 3 days/week (69 customers).
- Very few customers use the treadmill 6–7 days per week, indicating heavy users are a niche segment.
- Suggests products should cater primarily to moderate, regular usage rather than extreme daily usage.

Gender

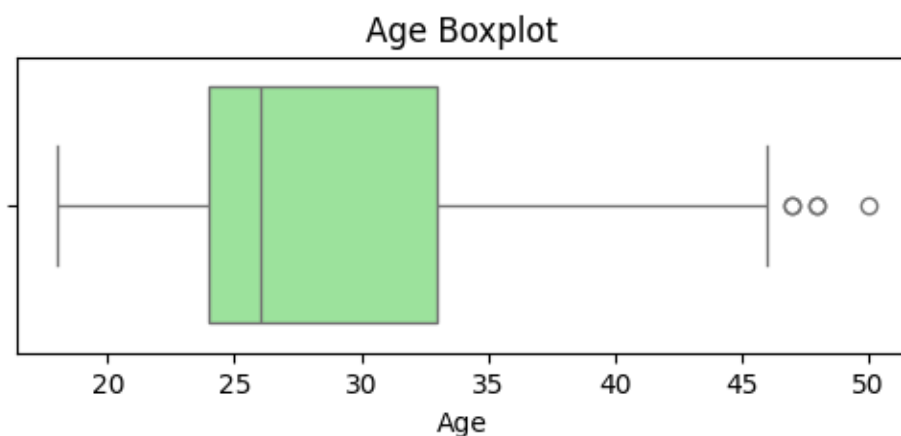
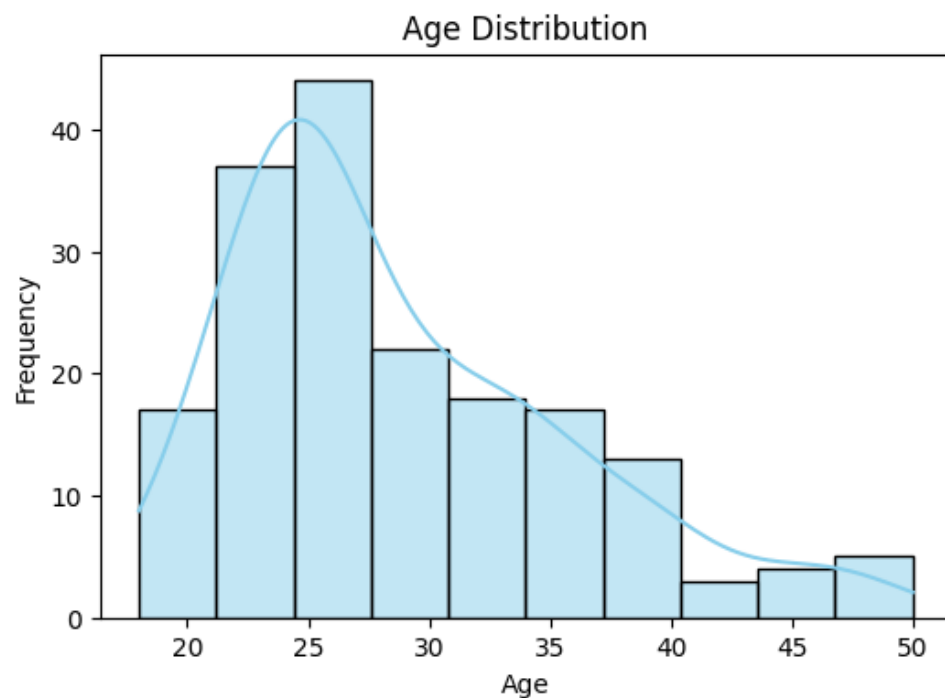
- Majority of customers are Male (104, 58%), while Females account for 42%.
- Slightly higher male representation may influence product marketing strategies.

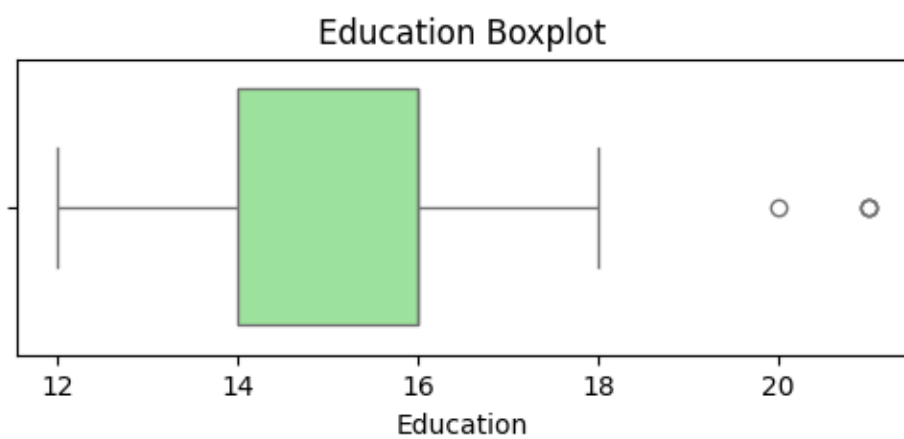
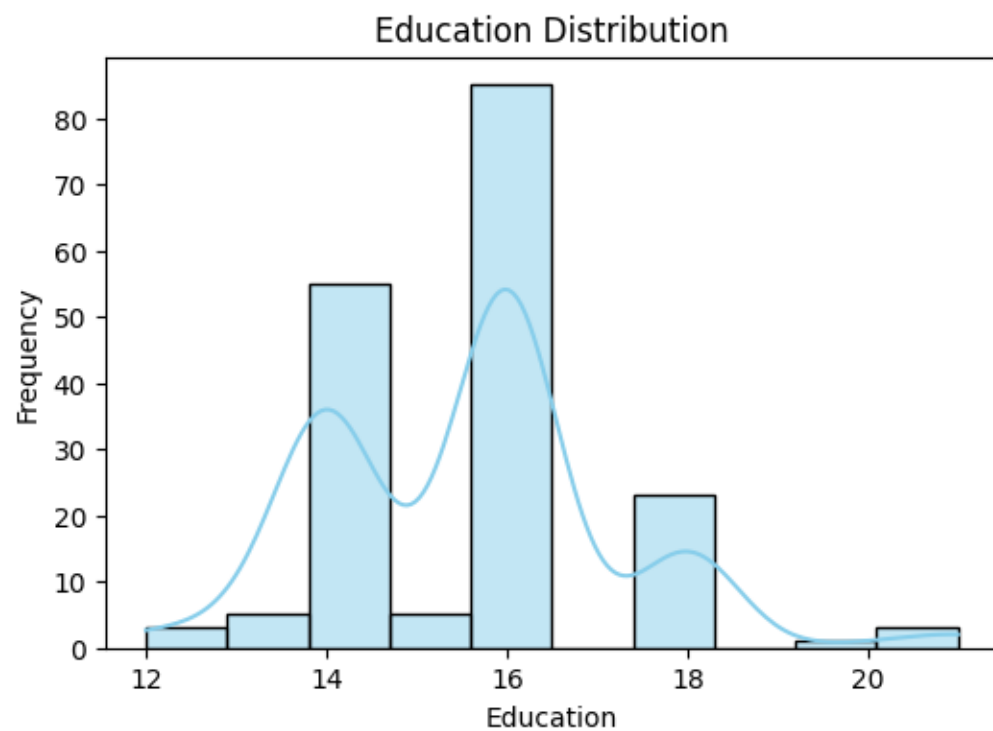
Marital Status

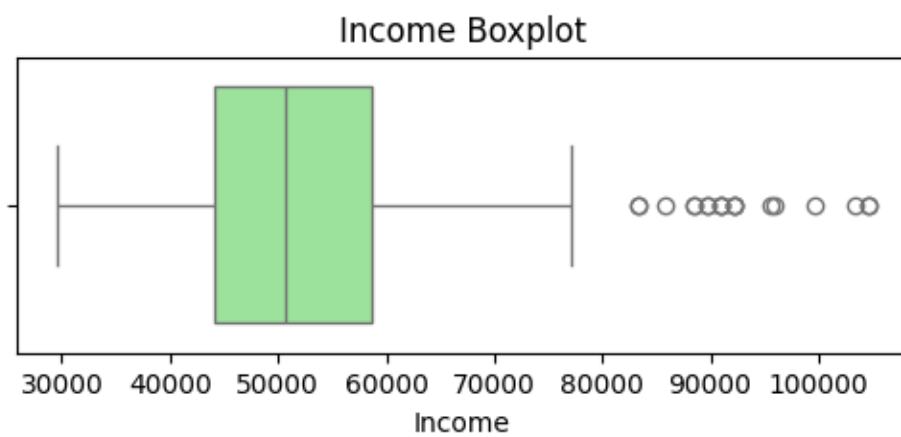
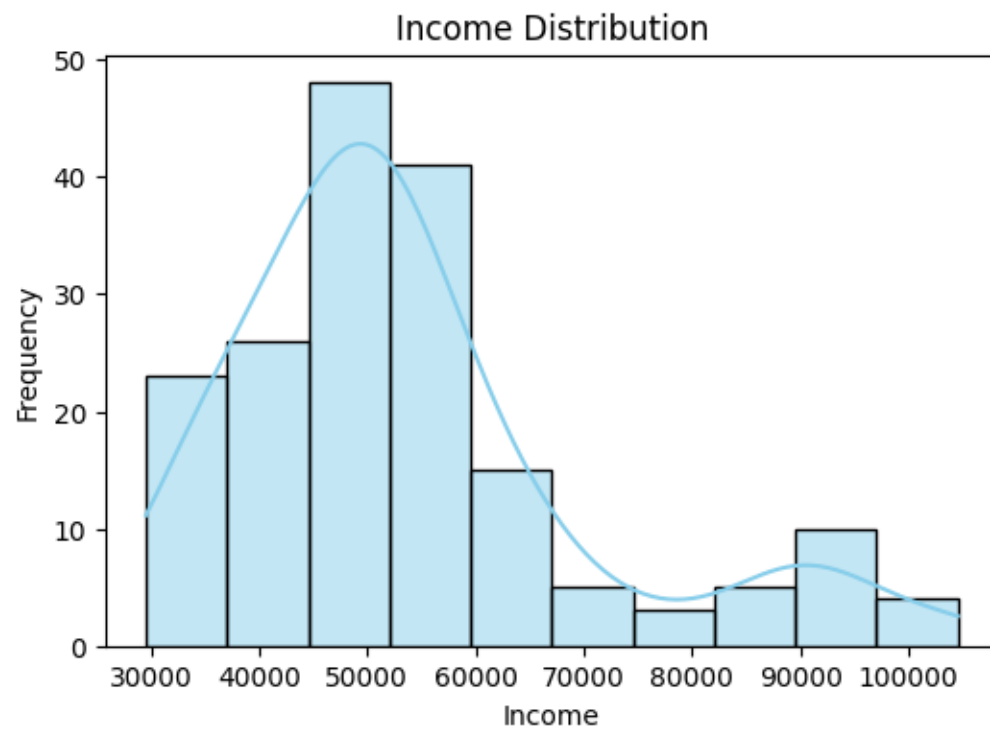
- Most customers are Partnered (107, 61%), compared to Singles (73, 39%).
- Suggests that treadmill purchases are more common among partnered individuals, possibly reflecting household investment in fitness.

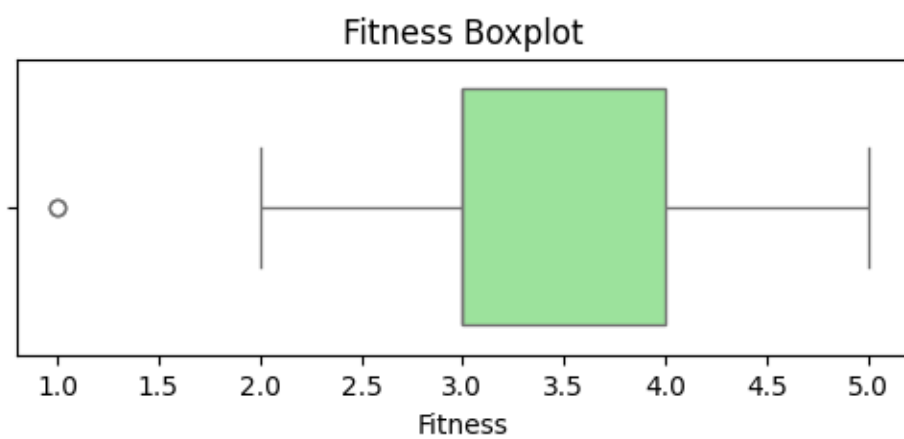
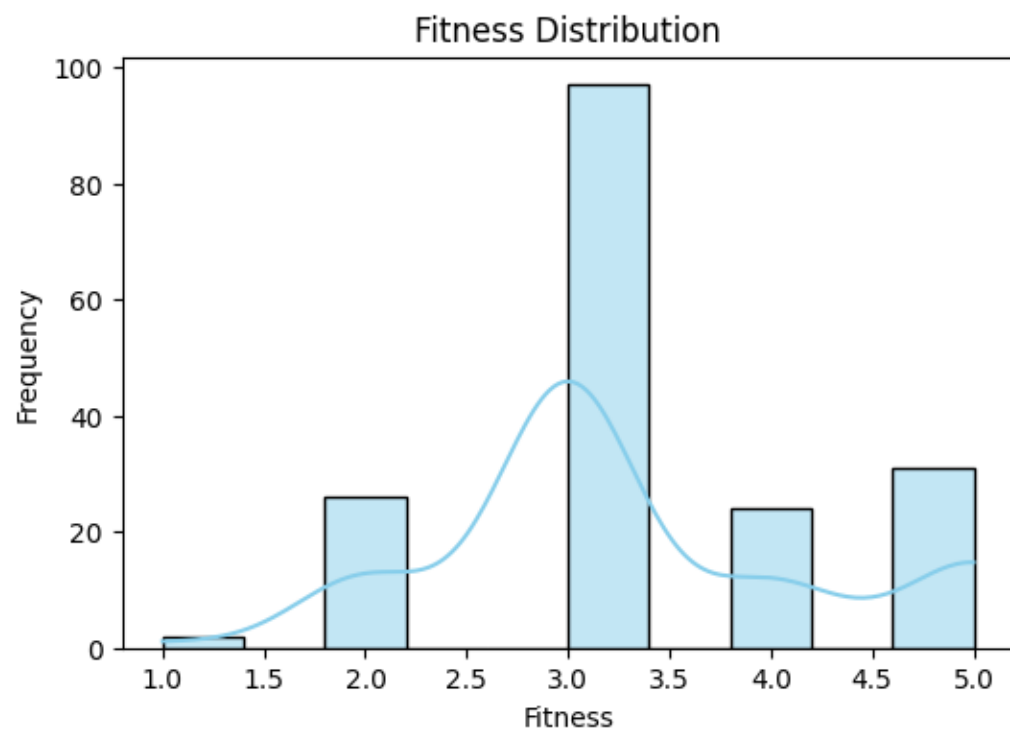
Age Group

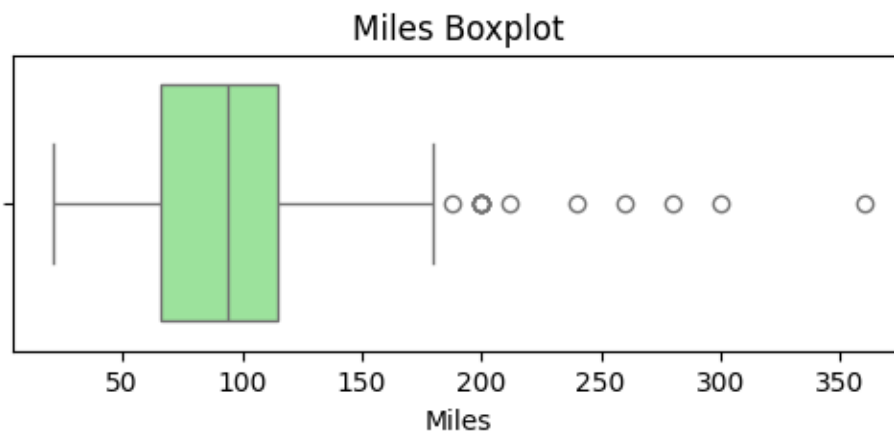
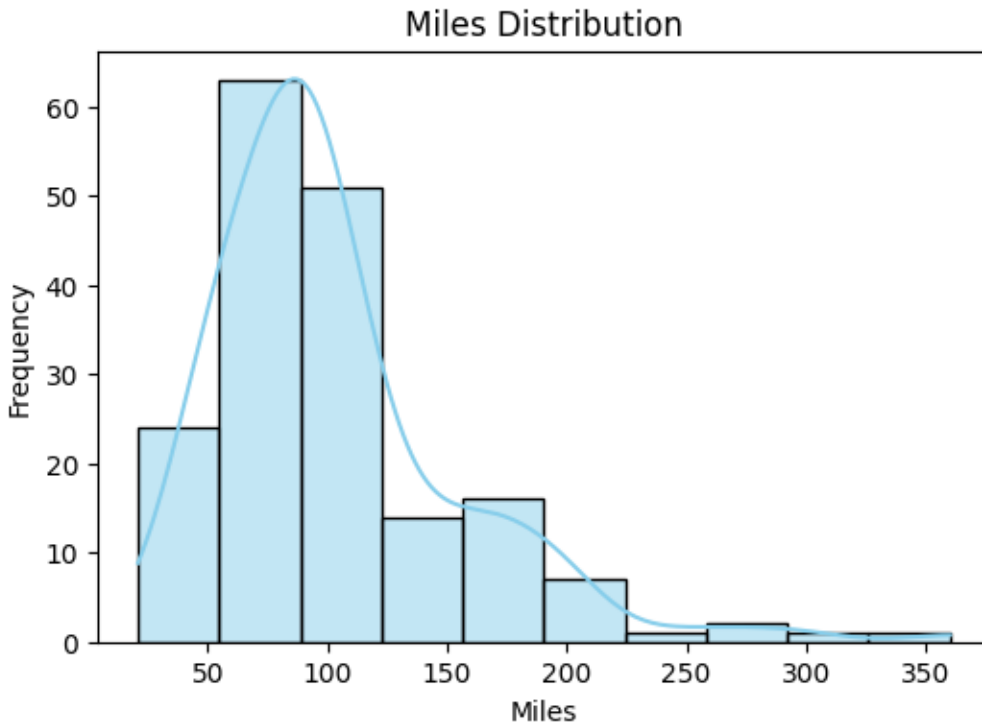
- Young Adults (18–25) and Adults (26–35) make up the majority (79 and 73 respectively), covering ~80% of customers.
- Middle-Age (36–45) and Older Adults (46–50) represent a smaller segment (~28%).
- Indicates the core target audience is young to mid-aged adults.











Univariate Analysis - Numerical Columns Quick Insights

Age

- Average age of customers is ~29 years, with a range from 18 to 50.
- Skewness = 0.98 → moderately positively skewed; more younger customers.
- Kurtosis = 0.41 → relatively flat distribution, indicating no extreme peaks.
- Most customers fall between 24–33 years (25th–75th percentile), highlighting the core young adult to adult segment.

Education

- Average education is ~15.6 years, mostly high school plus some college.
- Skewness = 0.62 → slightly positively skewed, a few customers with higher education levels.
- Kurtosis = 1.03 → distribution slightly peaked but fairly normal.
- Suggests customers are generally well-educated, which may influence marketing messages emphasizing fitness knowledge.

Income

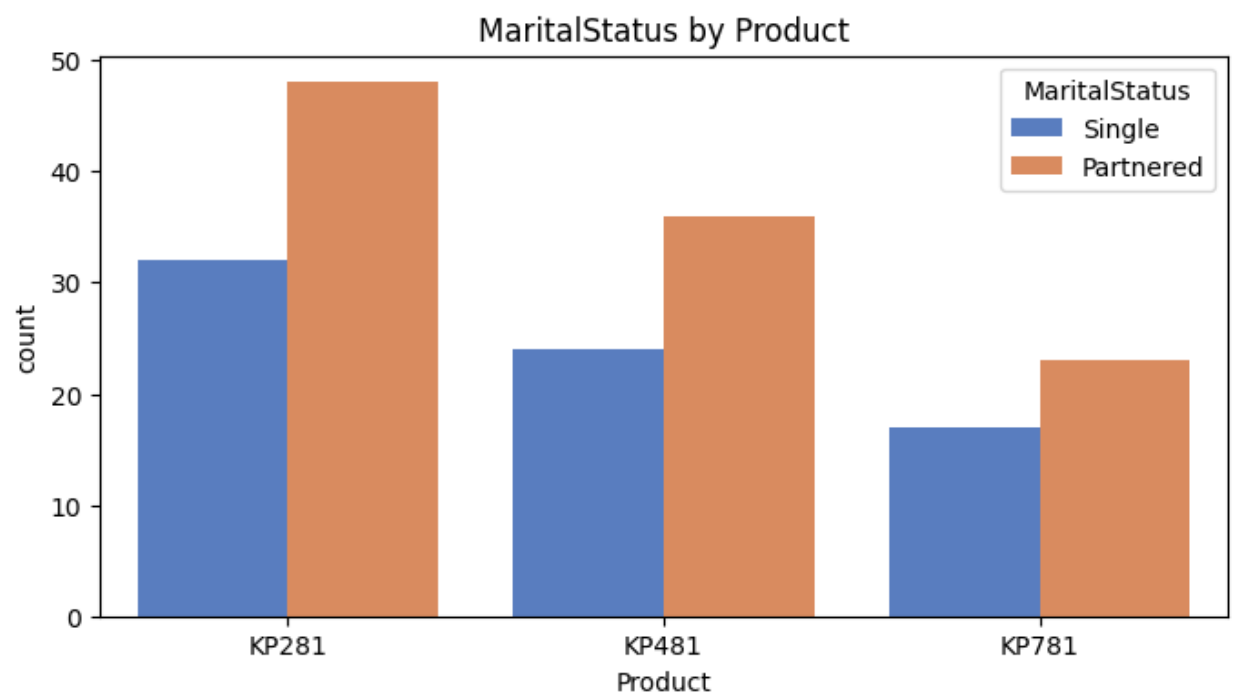
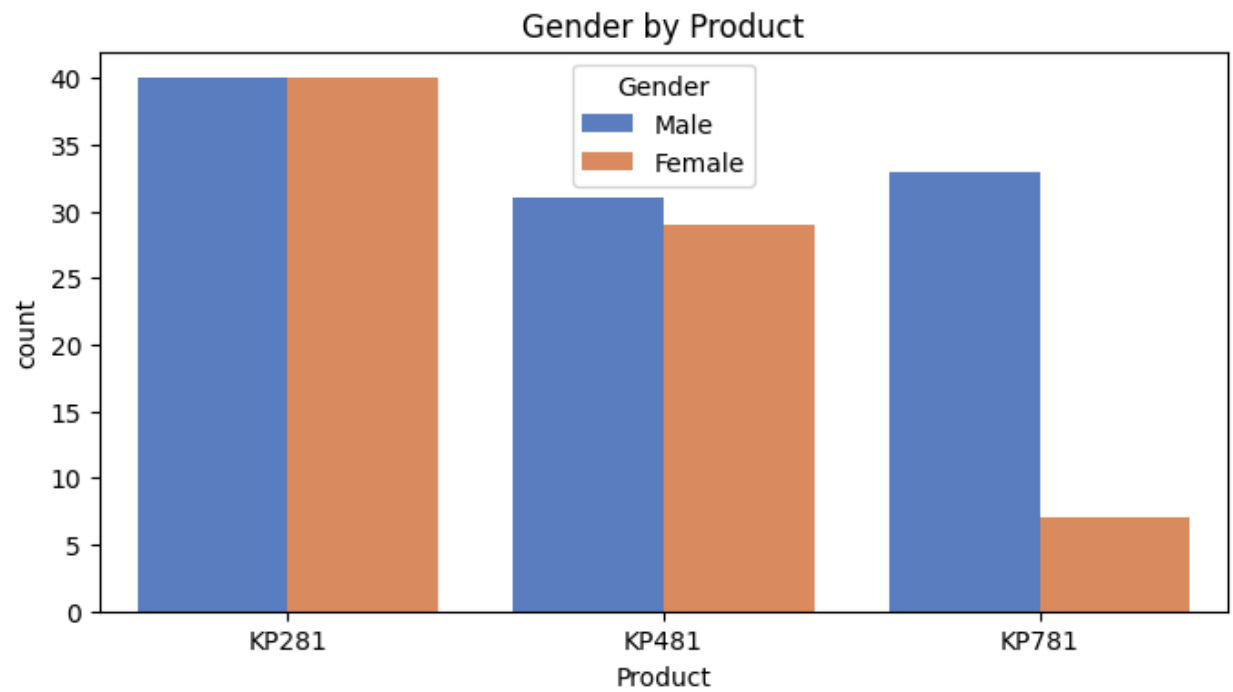
- Average income is ~53,700, ranging from 29,562 to 104,581.
- Skewness = 1.29 → positively skewed; most customers are in lower/middle-income range, few high-income outliers.
- Kurtosis = 1.37 → slightly heavy-tailed; some customers with exceptionally high income.
- 25th–75th percentile (44,059–58,668) confirms majority of customers fall in low to medium income, aligning with univariate categorical analysis.

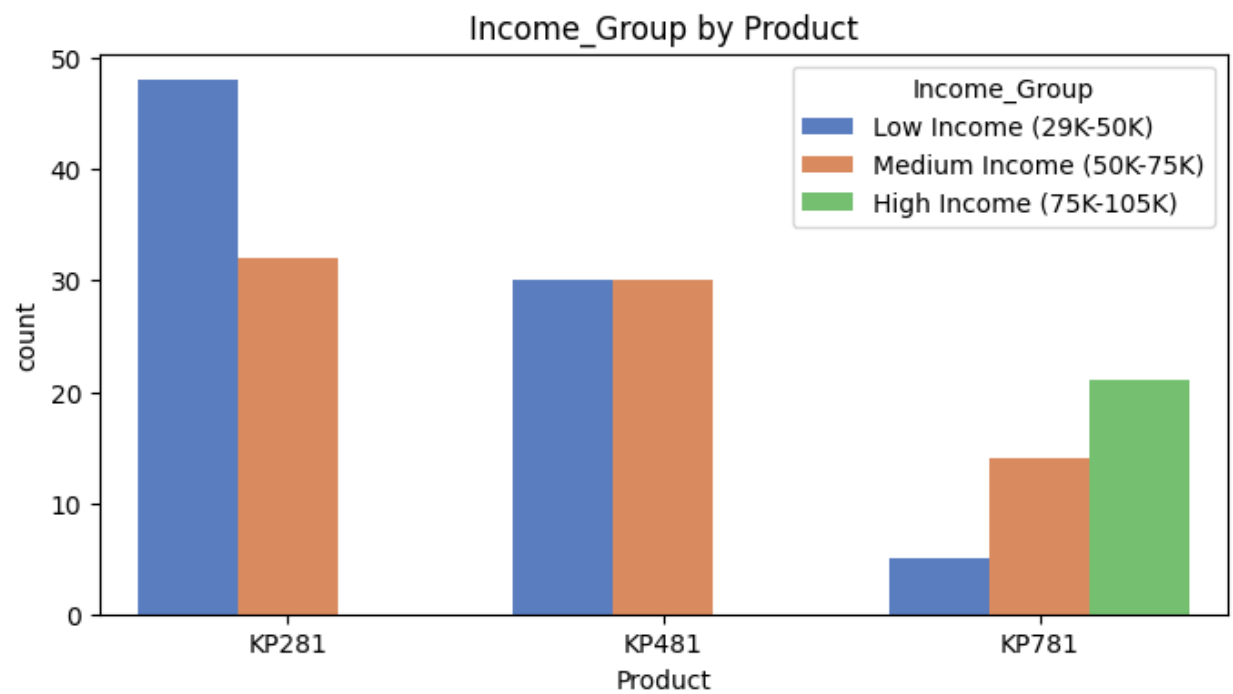
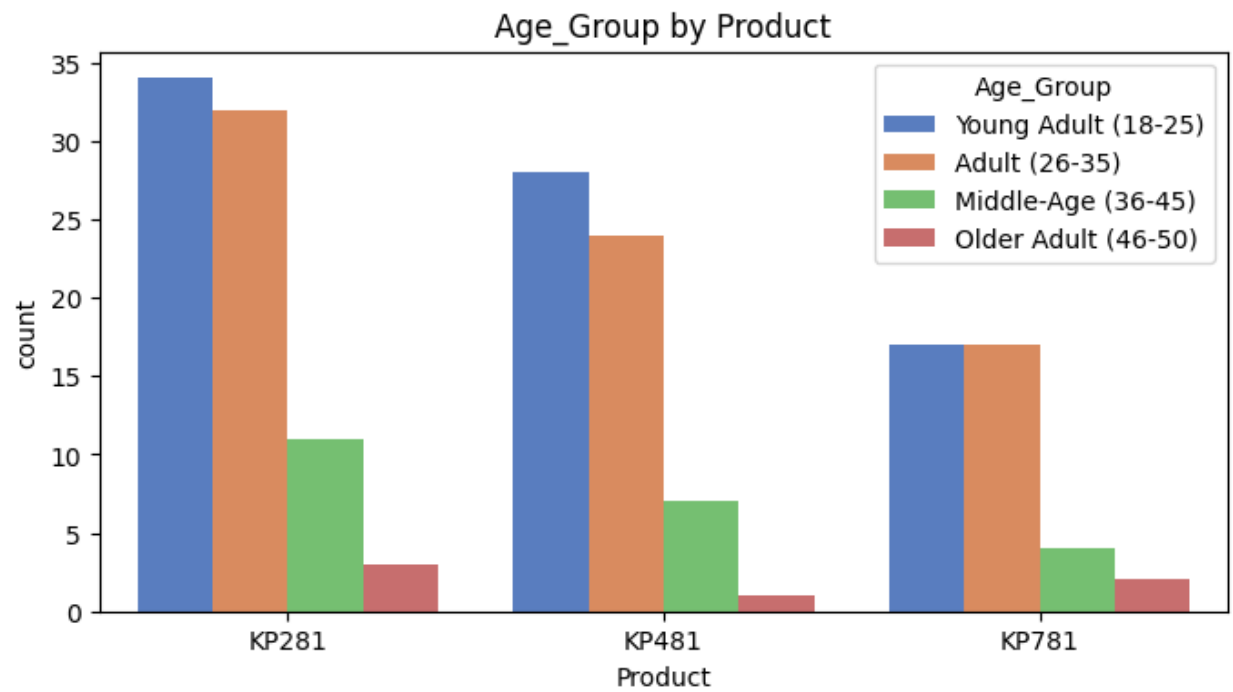
Fitness

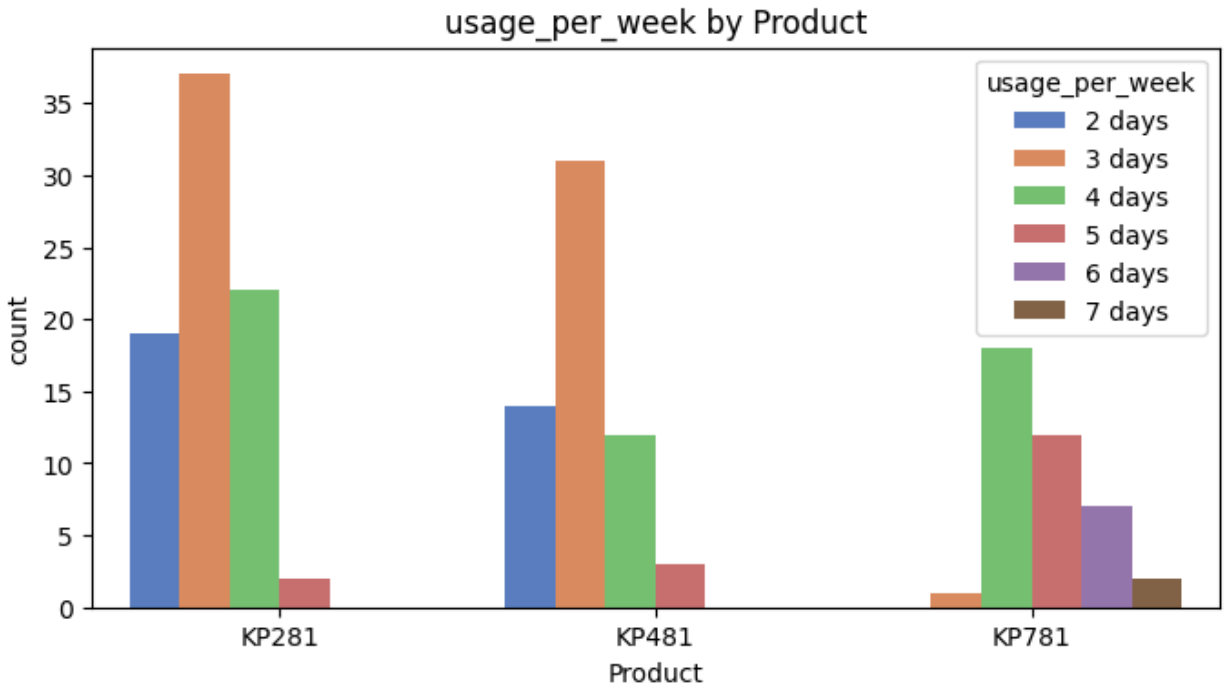
- Average fitness rating is 3.31/5, indicating moderate fitness.
- Skewness = 0.45 → slight positive skew; few customers report lower fitness.
- Kurtosis = -0.37 → slightly flat distribution; fitness ratings are spread across the scale.
- Most customers rate themselves 3–4, suggesting products appeal to moderately active users.

Miles

- Average weekly miles is ~103, with a large spread (21–360 miles).
- Skewness = 1.72 → strongly positively skewed; a few customers run significantly more than most.
- Kurtosis = 4.32 → leptokurtic; extreme values (high-mileage users) present.
- 25th–75th percentile (66–115 miles) indicates most customers run moderate distances, with high-mileage outliers forming a niche segment.







Bivariate Analysis - Categorical Columns Quick Insights

Gender vs Product

- KP281: Balanced gender distribution (50% Male, 50% Female).
- KP481: Slightly more males (52%) than females (48%).
- KP781: Majority male (82%), only 18% female. Business Insight:
- Entry-level KP281 appeals equally to both genders.
- Premium KP781 is predominantly purchased by males → marketing campaigns for KP781 could target male users more effectively.

Marital Status vs Product

- KP281 & KP481: 60% partnered, 40% single.
- KP781: Slightly lower proportion partnered (57%), 42% single. Business Insight:
- Most treadmill purchases are by partnered individuals across all products.
- Suggests campaigns emphasizing family/household fitness may resonate with the majority of users.

Age Group vs Product

- KP281: Mostly young adults (42%) and adults (40%).
- KP481: Young adults slightly higher (47%), adults 40%.
- KP781: Equal proportion of young adults and adults (42%), fewer middle-aged (10%), and some older adults (5%). Business Insight:

- All products target primarily 18–35 years, with KP281 slightly skewed younger.
- KP781 has slightly more adult users (26–35) compared to entry-level KP281 → product appeal shifts toward more established, active customers.

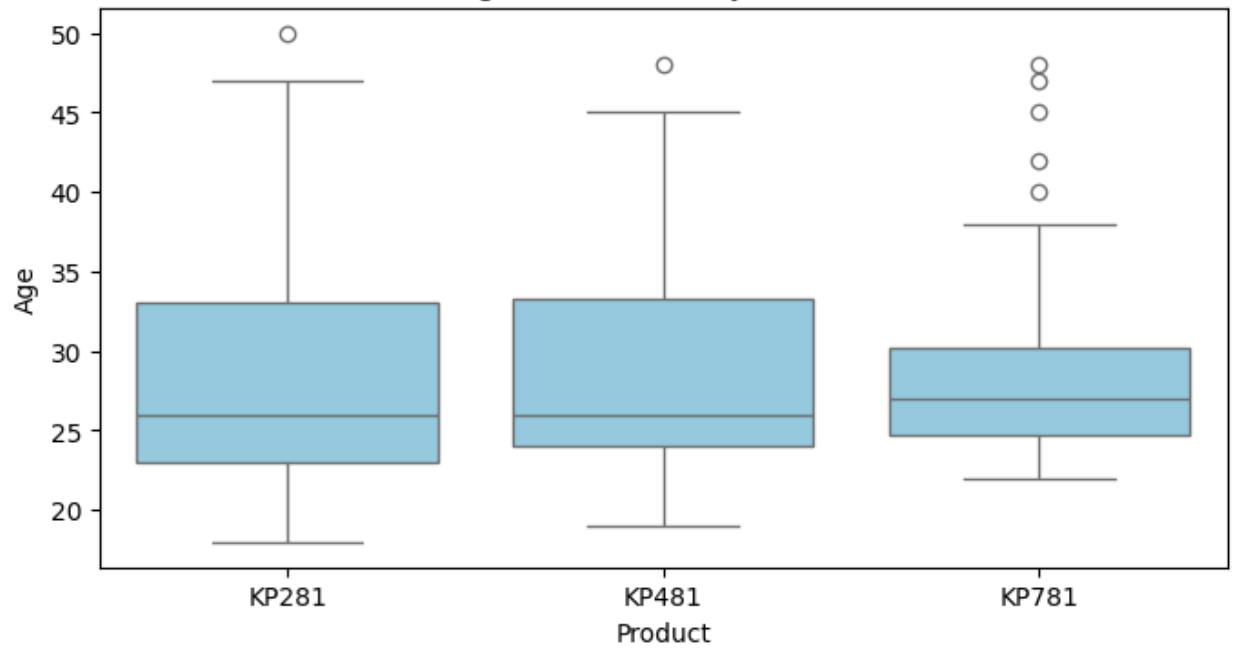
Income Group vs Product

- KP281: 60% low income, 40% medium income, 0% high income.
- KP481: Balanced between low (50%) and medium (50%), 0% high income.
- KP781: Majority high income (52%), some medium (35%), few low income (12%).
Business Insight:
- Entry-level and mid-level treadmills appeal to low-to-middle income groups.
- Premium KP781 is clearly purchased by high-income customers, aligning product tiers with income levels.
- Marketing messages can be tier-specific: affordability focus for KP281, premium features for KP781.

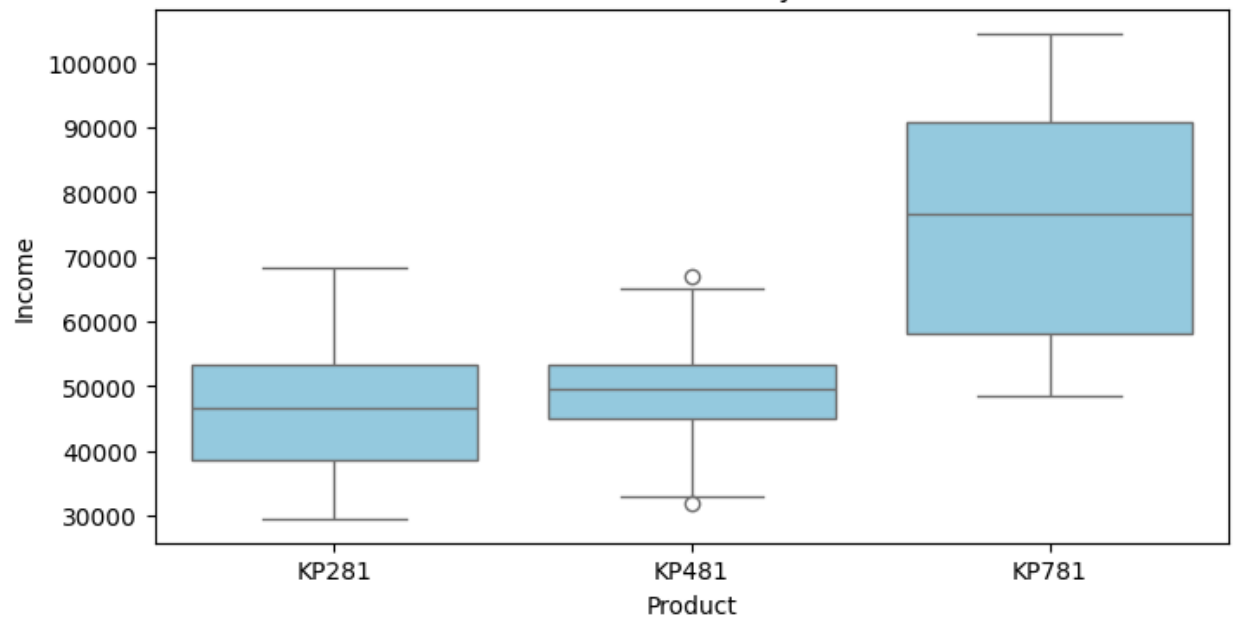
Usage per Week vs Product

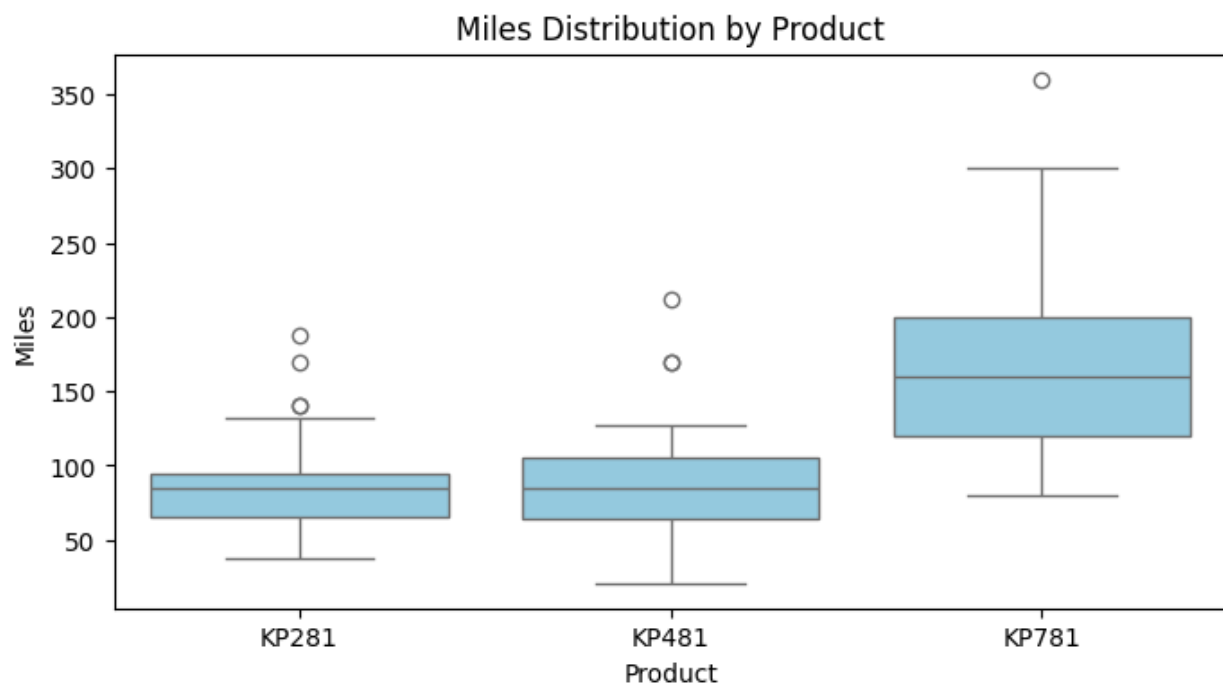
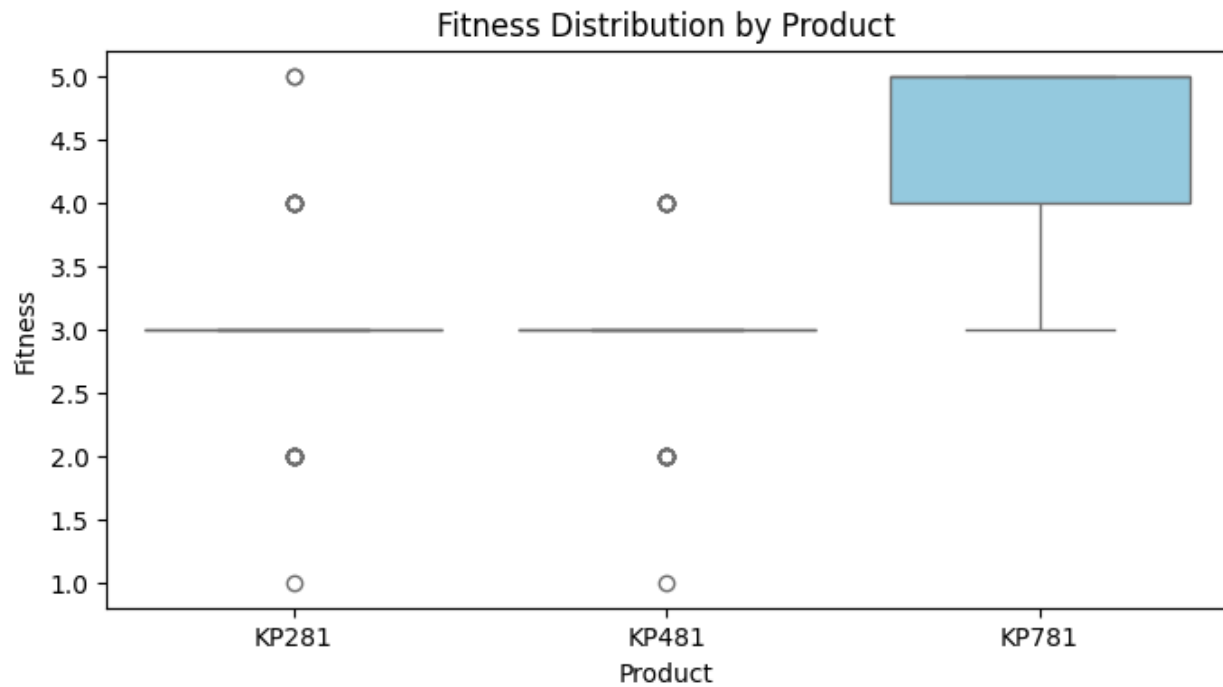
- KP281: 24% use 2 days/week, 46% use 3 days/week, 28% use 4 days/week, 2% use 5 days/week, 0% use 6–7 days/week.
- KP481: 23% use 2 days/week, 52% use 3 days/week, 20% use 4 days/week, 5% use 5 days/week, 0% use 6–7 days/week.
- KP781: 0% use 2 days/week, 2% use 3 days/week, 45% use 4 days/week, 30% use 5 days/week, 18% use 6 days/week, 5% use 7 days/week. Business Insight:
- Usage frequency aligns with product tiers: KP281 and KP481 appeal to casual/moderate users, while KP781 attracts high-frequency, dedicated users.
- Marketing can be tier-specific: convenience and accessibility for entry-level products, advanced performance and commitment for premium products.
- Usage patterns can guide personalized product recommendations for new customers based on expected weekly treadmill usage.

Age Distribution by Product



Income Distribution by Product





Bivariate Analysis - Numerical Columns Quick Insights

Age by Product

- KP281: mean 28.55, median 26, range 18–50
- KP481: mean 28.9, median 26, range 19–48

- KP781: mean 29.1, median 27, range 22–48 Insights:
- Age distribution is similar across products (core 26–29 years).
- KP781 slightly older on average, indicating premium users tend to be slightly more mature.
- Moderate variability across all products (range ~30 years).

Income by Product

- KP281: mean ~46k, median 46k, range 29.5–68k
- KP481: mean ~49k, median 49k, range 31.8–67k
- KP781: mean ~75k, median 76k, range 48.5–104k Insights:
- Clear income tier separation: KP781 is high-income users, KP281/KP481 are low-to-middle income.
- KP781 has much wider spread → some very high-income outliers.
- Boxplot comparison highlights product tier alignment with income.

Fitness by Product

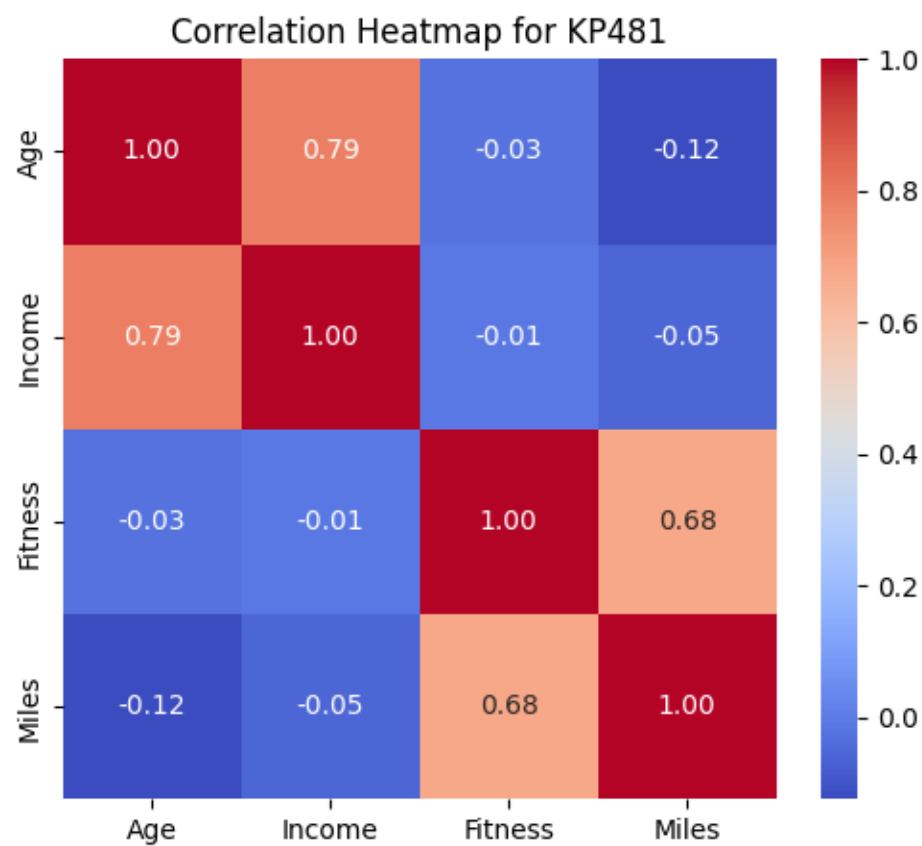
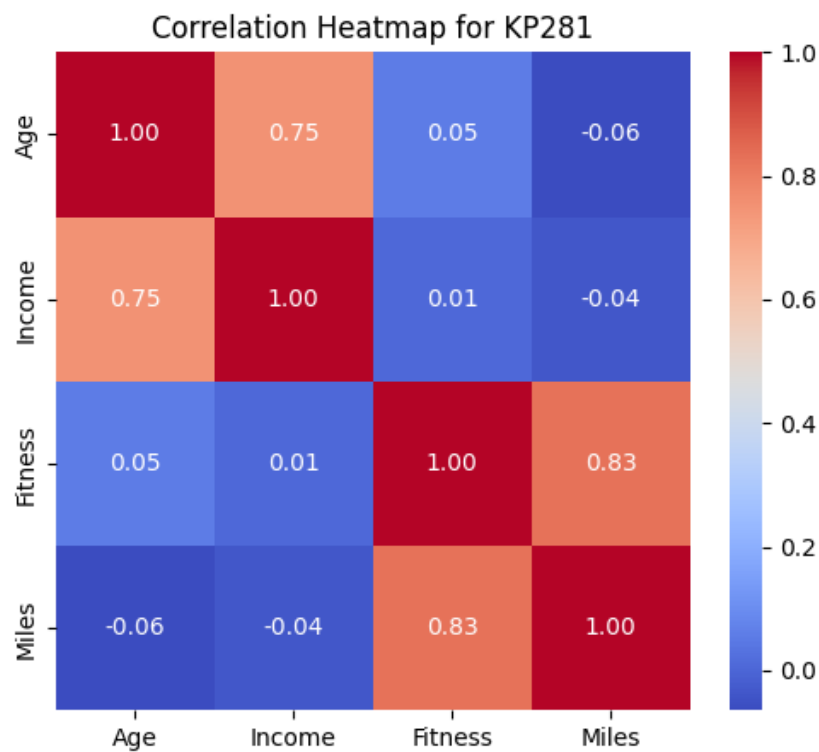
- KP281: mean ~2.96, median 3, range 1–5
- KP481: mean ~2.9, median 3, range 1–4
- KP781: mean ~4.63, median 5, range 3–5 Insights:
- KP781 users are significantly fitter than KP281/KP481 users.
- Median and mean differences indicate KP781 is right-shifted (higher fitness), confirming premium product attracts highly fit users.
- Low variability in KP781 (box smaller, high median) → consistently fit user segment.

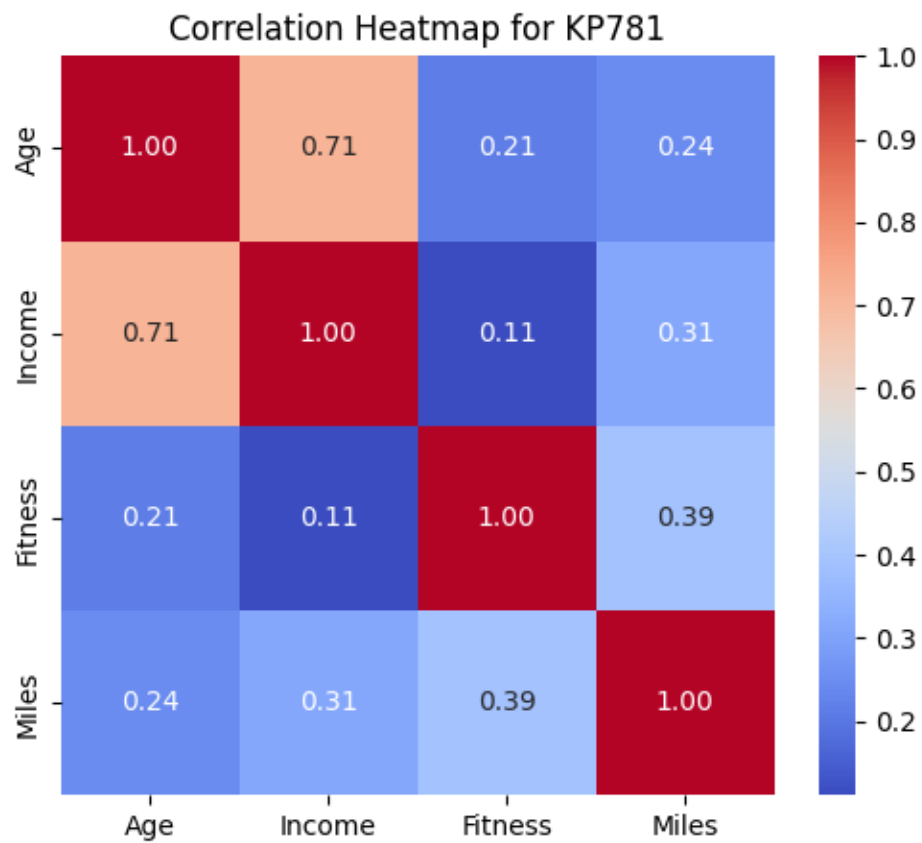
Miles by Product

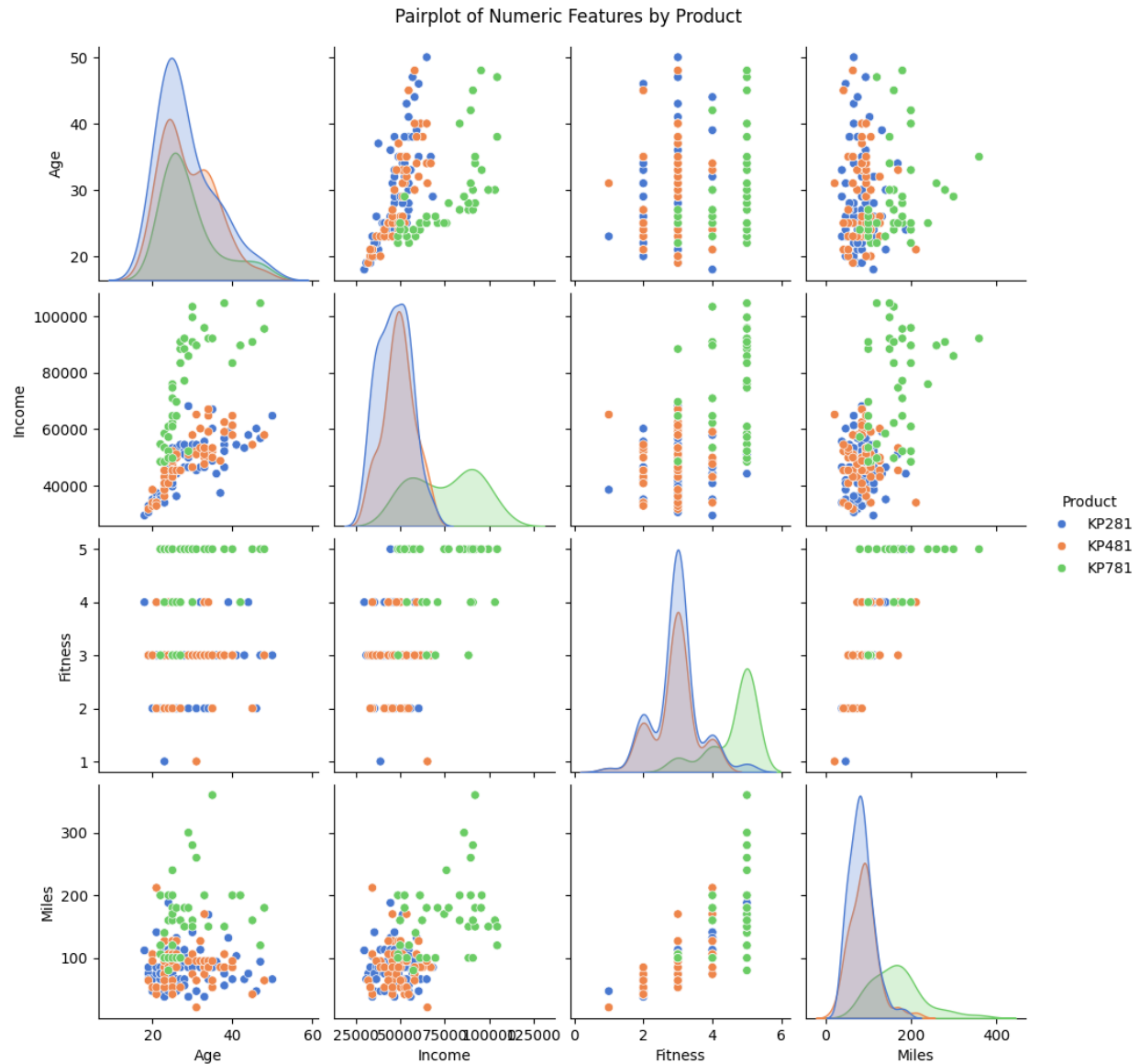
- KP281: mean ~83, median 85, range 38–188
- KP481: mean ~88, median 85, range 21–212
- KP781: mean ~167, median 160, range 80–360 Insights:
- KP781 users run much more miles than KP281/KP481 → heavy users cluster here.
- Larger range and higher median in KP781 → consistent with premium, highly active users.
- KP281/KP481 users have similar usage patterns (moderate miles), showing entry- and mid-level products target casual/moderate users.

Business Implications

- Premium KP781 targets high-income, highly fit, heavy users, slightly older than entry-level users.
- KP281 and KP481 appeal to moderate-income, moderately active users.







Correlation Analysis Quick Insights

Age vs Other Variables Age vs Income:

- Slight upward trend: older customers tend to earn a bit more, but the effect is moderate.
- Age vs Miles/Usage:
 - Younger users are scattered across low-to-moderate miles and usage.
 - No strong linear trend; age is less determinant of treadmill usage compared to fitness or product type.

Income vs Other Variables Income vs Miles:

- KP781 (green) cluster shows high-income, high-mileage users, clearly separated from KP281/KP481. Income vs Fitness:
- KP781 users are fitter, confirming premium product targets health-conscious, high-income segment.
- Insight: Income strongly aligns with product type and engagement level.

Fitness vs Miles

- Strong positive association across products: fitter users cover more miles.
- KP781 users are distinctly clustered in high fitness, high miles, reinforcing product differentiation.

Product Clustering

- KP281 (blue): Moderate income, moderate fitness, lower miles.
- KP481 (orange): Slightly higher income than KP281, moderate fitness, moderate miles.
- KP781 (green): High income, very fit, heavy mileage users.

Heatmap Insights

- Miles vs Usage: Strong positive correlation - high-frequency users run more miles.
- Fitness vs Miles: Positive correlation - fitter users engage more in activity.
- Income vs Miles/Fitness: Moderate positive correlation - high-income users tend to be fitter and more active.
- Age correlations: Weak correlations with most variables - age is less predictive than income or fitness for product segmentation.

Business Insights

- Miles and Usage: engagement
- Fitness: readiness for premium product
- Income: purchasing power

Customer Profiling for Each Product

1. KP281 – Entry-Level Treadmill

Price: \$1,500

Demographics:

- Gender: Balanced (50% Male, 50% Female) → appeals equally to both genders.

- Marital Status: Mostly partnered (60%), 40% single → suitable for household/family-oriented campaigns.
- Age Group: Primarily young adults (42%) and adults (40%), mean age ~28.5 → slightly skewed toward younger users.
- Income Group: 60% low income, 40% medium income → budget-conscious, affordability-focused segment.

Behavior & Usage:

- Usage Frequency: Casual/moderate users (2–4 days/week; median 3 days).
- Fitness Level: Moderate fitness (mean ~3/5).
- Miles Run: Moderate mileage (mean ~83 miles).

Insights / Marketing Persona:

- Entry-level, price-sensitive customers who want basic treadmill functionality for casual home fitness.
- Likely to be younger, partnered individuals starting or maintaining a fitness routine.

Marketing message: affordability, convenience, simple features, and family/household fitness appeal.

2. KP481 – Mid-Level Treadmill

Price: \$1,750

Demographics:

- Gender: Slightly more males (52%) than females (48%).
- Marital Status: Mostly partnered (60%).
- Age Group: Young adults (47%), adults (40%), mean age ~28.9 → similar to KP281, slightly more male.
- Income Group: Balanced between low and medium income (50% each) → middle-income aspirational buyers.

Behavior & Usage:

- Usage Frequency: Moderate users (2–5 days/week; median 3 days).
- Fitness Level: Moderate fitness (mean ~2.9/5).
- Miles Run: Moderate mileage (mean ~88 miles).

Insights / Marketing Persona:

- Customers seeking a step-up from basic treadmills—slightly more serious about fitness but not extreme athletes.

- Likely to value durability, a few advanced features, and moderate performance.

Marketing message: reliable mid-tier product for regular home workouts, bridging affordability and performance.

3. KP781 – Premium Treadmill

Price: \$2,500

Demographics:

- Gender: Predominantly male (82%), 18% female → campaigns can target male users more.
- Marital Status: 57% partnered, 42% single → appeals to both individual and household buyers.
- Age Group: Equal proportion of young adults and adults (42%), fewer middle-aged (10%), mean age ~29.1 → slightly older than entry-level buyers.
- Income Group: Majority high income (52%), some medium (35%), few low income (12%) → high purchasing power segment.

Behavior & Usage:

- Usage Frequency: Heavy, committed users (4–7 days/week; median 5 days).
- Fitness Level: Very fit (mean ~4.63/5) → consistently fit, high-performing users.
- Miles Run: High mileage (mean ~167 miles) → premium product attracts serious runners.

Insights / Marketing Persona:

- Premium, high-income, highly fit users with a strong commitment to fitness.
- Likely to use the treadmill intensively, value advanced features, and expect high performance.

Marketing message: elite performance, advanced technology, heavy usage capability, and fitness optimization.

Customer Profiling per Product - Summary Table								
Product	Price	Gender	Age Group	Income	Fitness	Usage (days/week)	Miles	Customer Persona
KP281	\$1,500	50M/50F	18–29	Low/Medium	Moderate	2–4	83	Casual, younger, budget-conscious, household fitness starter
KP481	\$1,750	52M/48F	19–29	Low/Medium	Moderate	2–5	88	Regular, moderately serious, value-conscious fitness enthusiasts
KP781	\$2,500	82M/18F	22–35	High	Very Fit	4–7	167	Dedicated, high-income, performance-driven, elite fitness users